

Pathogenicity of three formulations of *Beauveria bassiana* and efficacy of autoinoculation devices and sterile fruit fly males for dissemination of conidia for the control of *Ceratitis capitata*

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Abstract

This study reports the pathogenicity of three formulations of *Beauveria bassiana* (Balsamo) Vuillemin (Hypocreales) – Bb-ET, GHA, and Bb-AES – and their application in panel-type and cylinder-type autoinoculation devices, and using sterile males as vectors for the control of *Ceratitis capitata* (Wiedemann) (Diptera: Tephritidae) in coffee-producing areas of the Central Highlands of Guatemala. Using sterile adults of *C. capitata* under laboratory bioassays, mean lethal concentrations (LC₅₀) of 1.4×10^7 , 2.4×10^6 , and 8.2×10^5 conidia ml⁻¹, and median survival times of 4.1 ± 0.1 , 4.2 ± 0.1 , and 3.8 ± 0.1 days, were recorded for the strains GHA, Bb-AES, and Bb-ET, respectively. These values indicate that the three strains were sufficiently pathogenic and that their period of biological activity was considered adequate for using sterile flies as vectors of the inoculum. Observed percentages of sporulated wild *C. capitata* flies were 57.3, 44.7, and 44.3% for sterile fly vectors, panel devices, and cylinder devices, respectively. The total population reduction at the end of the study period was over 90% for the three treatments. Our results show that sterile males used as vectors or disseminator devices may represent a new control method for area-wide integrated management of Mediterranean fruit flies. The potential of this new approach and its integration with other control methods, including the sterile insect technique, is discussed.

Introduction

Biological pest control using entomopathogenic fungi has been consolidated as an effective and viable strategy in organic and conventional agricultural systems, such as horticulture, coffee, sugarcane, and fruit growing (Dolin-ski & Lacey, 2007). Laboratory and field evaluations with

Beauveria bassiana (Balsamo) Vuillemin and *Metarhizium anisopliae* (Metschnikoff) Sorokin (Hypocreales) indicate that the use of either entomopathogen is a potential alternative for fruit fly management (De la Rosa et al., 2002; Ekesi et al., 2002; Quesada-Moraga et al., 2008; Toledo et al., 2008; Muñoz et al., 2009; Novelo-Rincón et al., 2009). To achieve greater effectiveness of this pest control method, the strains must be characterized, and the most suitable way to apply the fungi must be determined. This may involve the use of water as a vehicle, or the use of dry conidia in autoinoculation devices or by using live insects as vectors (Vega et al., 2000; Maniania, 2002). The viability

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and capacity of pathogen transmission determines the range of propagation in the target host (Vega et al., 2000; Toledo et al., 2007; Quesada-Moraga et al., 2008).

Dissemination of entomopathogenic microorganisms with autoinoculation devices and the use of insect vectors has been demonstrated (Vega et al., 1995; Toledo et al., 2007; Migiro et al., 2010; Flores et al., 2013). Autoinoculation devices attract and contaminate the pest insect, which subsequently goes on to disseminate the conidia among the wild population (Maniania, 2002; Dimbi et al., 2003; Vickers et al., 2004). The advantages of using autoinoculation devices include: (1) greater protection of the conidia against radiation, longer retention of conidial viability (Maniania, 2002); (2) increased specificity when a specific attractant to the target species is incorporated; (3) smaller quantities of inoculum are used; and (4) infected insects disperse the pathogen during interaction with other individuals of their species (Vega et al., 2000). When using sterile insects as vectors, it will be expected that the inoculum will directly contact a higher number of wild conspecifics. If mating does not occur, there might be horizontal transmission that will result in the death of the wild female, and fungal infection will prevent a wild female to remate with a wild male and reproduce. In areas with difficult access, sterile males can find and interact with the wild population transmitting the pathogen.

The Mediterranean fruit fly, *Ceratitis capitata* (Wiedemann) (Diptera: Tephritidae), is a pest of African origin that has spread across the world, colonizing important tropical and subtropical areas (Gutiérrez-Samperio, 1976). The countries that are free of this pest prevent its introduction and spread by applying exclusion actions such as trapping for early detection and emergency response (Enkerlin et al., 2015). In Guatemala, *C. capitata* is controlled with an area-wide integrated pest management (IPM) approach including the use of the sterile insect technique (SIT) and selective ground or aerial spraying of GF-120 NF Naturalyte bait (Villaseñor & Ortiz, 2010). These control actions have contained the advance of the pest toward Mexico and the pest-free regions of Guatemala and Belize (Villaseñor et al., 2000; Hendrichs et al., 2002; Enkerlin et al., 2015). However, it is still important to develop effective complementary control alternatives with an emphasis on lower environmental impact and greater acceptance by fruit and coffee growers, especially in organic agricultural systems. In this context, it is considered that the fungus *B. bassiana* can represent an additional control method in the system.

We present the results of bioassays to characterize the pathogenicity and virulence [median survival time (MST) and mean lethal concentration (LC_{50})] of three commercially available *B. bassiana* conidia products against *C. capitata*. We also compare the use of sterile males as vectors of *B. bassiana* and the effectiveness of two devices developed to disseminate conidia of this fungus.

Materials and methods

Biological material

The strains of *B. bassiana* evaluated were as follows: (1) GHA, a commercial formulation named Botanigard 22 WP in inert material and components derived from petroleum with 4.4×10^{10} conidia g^{-1} (BioWorks, Victor, NY, USA); (2) Bb-ET, formulated with celite powder 400 (diatomaceous earth, Hyflo; Celite, Lompoc, CA, USA) at a concentration of 2.9×10^9 conidia g^{-1} (Laboratorio de Organismos Benéficos, Tapachula, Chiapas; registration SAGARPA-DGSV 05-707-001-CB), and (3) Bb-AES (Agricultura El Sol from Guatemala City) with 7.2×10^9 conidia g^{-1} , the inoculum of this strain was produced in Potato Dextrose Agar (PDA) (65 g agar + 1 l distilled water, and 5 g yeast extract). The conidia of the previously described strain were collected by scraping the surface of the culture medium with a sterile inoculation loop, and the material was then deposited in amber-colored plastic containers, which were stored at 4 °C until subsequent use in the bioassays. Humidity inside the container was 60% r.h.

Conidial viability of the fungus was determined before proceeding to the pathogenicity bioassay, according to the microculture technique (Jiménez, 1992) and by observations counting the germinated conidia at 40× with a compound microscope (Dialux 20 EB; Leitz, Wetzlar, Germany). Conidia were considered to have germinated when the germ tube was longer than the diameter of the conidium. All strains used had >90% germination.

For the laboratory bioassays, 5 000 pupae of *C. capitata* (strain Vienna 8), dyed with pink fluorescent Day-Glo colorant (Color Corporation, Cleveland, OH, USA), and irradiated with Cobalt-60 at 125 Gy (Gómez-Símuta, 2010) were provided by the Laboratorio de Cría y Esterilización of the Programa Moscamed, in Metapa de Domínguez, Chiapas, Mexico. The pupae were distributed in five glass cages (30 × 30 × 30 cm), and the emerged sterile males were fed with a mixture of hydrolyzed protein (MP Biomedicals, Santa Ana, CA, USA) and sugar (1:3 ratio). Water was supplied in vials covered with cotton wool swabs. The flies in the cages were kept in the laboratory at 26 ± 1 °C, $75 \pm 5\%$ r.h., and L12:D12 photoperiod for 4 days prior to conducting the bioassays.

Median survival time (MST)

For each strain, four groups of 15 males were placed in test tubes (18 × 150 mm) containing 2 ml of 1% spores suspension of each commercial product in a solution of 0.02% of Tween-80, agitating softly for 30 s. The Bb-AES strain was directly suspended in distilled water. Treated flies were placed on a paper filter to eliminate excess of moisture and transferred to glass cages (30 × 30 × 30 cm), provided with food and water, as described above. Daily, over a period of 15 days, mortality was checked and the dead males were placed in humid chambers in order to induce mycelium growth. After 5 days, dead flies were inspected under a stereoscopic microscope (Carl Zeiss, Göttingen, Germany) to confirm the presence of *B. bassiana* mycelium. Five replicates were conducted per strain, each consisting of five glass cages containing 60 flies per cage.

Mean lethal concentration (LC₅₀)

For each strain, from a 1% spore suspension, six dilutions were prepared + a control (water only) for the bioassays (Table 1). For each concentration of each strain, 100 treated males were inoculated and placed in glass cages as described above. Mortality was recorded daily, and the dead flies were checked to verify fungal infection as previously described. Each strain was replicated 4×, each replicate started with a different batch of pupae and consisted of a glass cage with 100 adults per concentration.

Study sites for evaluating the conidial autoinoculation devices

The study was conducted from February to July 2012 in the coffee-producing zone of the Sacatepéquez Department, in Guatemala, at sites located at 1 300–1 500 m above sea level. Mean temperature was 18 °C (range 8–22 °C), and annual precipitation was 970–1 777 mm (ANACAFE, 2012). The area presented coffee plantations with infestations of the Mediterranean fruit fly, where the

Regional Moscamed Program Mexico-Guatemala-USA conducted control actions based on the concept of area-wide IPM (i.e., SIT, bait stations, and ground spraying of GF-120) (PRMGMEU, 2009). To evaluate the infection induced by *B. bassiana* on wild populations of *C. capitata*, three plots of 100 ha each were randomly assigned for each treatment (i.e., 1 200 ha in total in the experimental area). Each plot was considered an experimental unit. Four treatments were evaluated: (1) sterile males as vectors, (2) cylinder autoinoculative devices, (3) panel autoinoculative devices, and (4) untreated control. Each treatment was replicated 3×. For this phase we selected the Bb-ET strain, which had the lowest LC₅₀.

Release of sterile male vectors. To evaluate conidial transmission in the field, 1.5 million pupae dyed with pink fluorescent colorant and irradiated with ⁶⁰Co at 125 Gy were supplied each week by the El Pino facility of the Regional Moscamed Program in Guatemala. For emergence and sexual maturation, these pupae were distributed in batches of 50 000 and placed in PARC boxes (60 × 33 × 48 cm Plastic Adult Rearing Container, at 23.5 ± 5.5 °C and 75.5 ± 5.5% r.h.) with food and water provided ad libitum. The boxes were maintained in a room at 23.2 °C and 65 ± 5% r.h. (Zavala-López & Enkerlin, 2016). When the flies reached 6 days of age, the ventilation windows of the PARC boxes were sealed and inoculation was carried out (Figure 1A) applying 2.5 g of conidia per box with a dust sprayer (Guarany Industria e Comercio, Sao Pablo, Brazil). The males were treated with a ratio of 0.05 g of conidia per 1 000 specimens (1.45 × 10⁵ conidia per fly). Inoculated males were transferred to the target area for ground release at a density of 4 800 adults ha⁻¹ (= 0.24 g of conidia ha⁻¹). A total of 23 weekly releases were conducted.

Cylindrical device. A 500-ml polyethylene terephthalate (PET) container (14.0 cm high, 8.5 cm diameter) was used, with two diametrically opposed holes of 3 cm diameter in the upper third of the container. The lid had four triangular apertures of 2.25 cm² and the container bottom was also open. The lid and bottom of the devices were covered with tulle cloth in order to prevent the entry of flies (Figure 1B). Inside each device, a small basket containing a 3-g plug of trimedlure was hung from the lid. The device was covered with yellow villous cloth (14 × 22 cm). Aliquots of 2 g of *B. bassiana* conidia were applied to the external surface of this cloth with a 5-cm brush. The cloth was replaced every 15 days with another recently impregnated with conidia. In each 100-ha plot, a density of one device ha⁻¹ was maintained.

Table 1 Concentrations of conidia ml⁻¹ of three commercial products of *Beauveria bassiana* used to calculate the lethal concentration (LC₅₀) for sterile males of *Ceratitidis capitata*

Fungal suspension (%)	Bb-Et	GHA-Botanigard 22WP	Bb-AES
0.01	2.9 × 10 ⁵	4.4 × 10 ⁷	7.2 × 10 ⁵
0.03	8.7 × 10 ⁵	1.3 × 10 ⁸	2.2 × 10 ⁶
0.05	1.5 × 10 ⁶	2.2 × 10 ⁸	3.6 × 10 ⁶
0.1	2.9 × 10 ⁶	4.4 × 10 ⁸	7.2 × 10 ⁶
0.3	8.7 × 10 ⁶	1.3 × 10 ⁹	2.2 × 10 ⁷
0.5	1.5 × 10 ⁷	2.2 × 10 ⁹	3.6 × 10 ⁷
1	2.9 × 10 ⁷	4.4 × 10 ⁹	7.2 × 10 ⁷

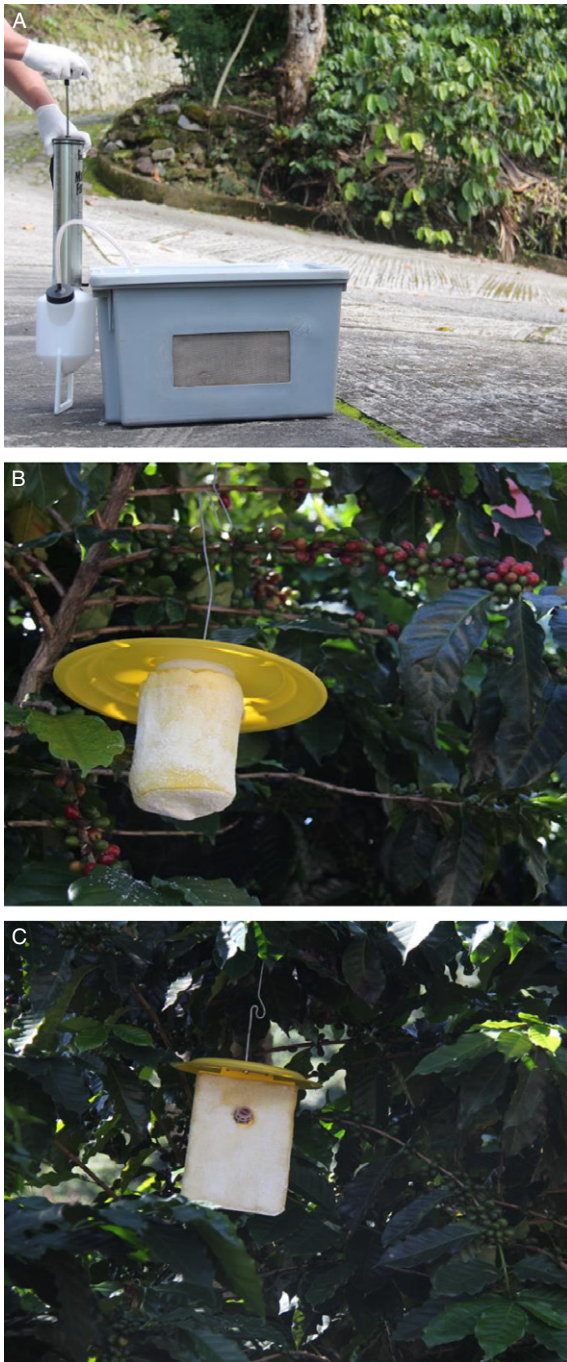


Figure 1 (A) Plastic adult rearing container (PARC) box and dust sprayer to inoculate the flies and (B) cylindrical, and panel (C) devices inoculated with *Beauveria bassiana* conidia to control wild Mediterranean fruit fly populations.

Panel device. This was made from a galvanized steel sheet of 23×14 cm with a 2.5-cm-diameter orifice in the center, where a small plastic basket with the 3-g trimmed lure

plug was hung. This device was covered with yellow villous cloth impregnated with 2 g of conidia (Figure 1C). The cloth was replaced every 15 days with another recently impregnated with conidia and, in each 100-ha plot, one device ha^{-1} was maintained.

Untreated control. No *B. bassiana* was applied in the area designated as the untreated control.

In order to monitor the wild population in the four treatments, one Phase IV EXA[®] trap (Better Word Manufacturing, Fresno, CA, USA) baited with BioLure Unipack (Suterra/AgriSence, Glamorgan, UK) was installed for each 4 ha. The activity of this attractant lasts 120 days under field conditions (Peñarrubia-María et al., 2014). The traps were checked each week and the inserts with insects were transferred for identification to the laboratory of the Centro de Operaciones del Altiplano Central of the Moscamed Program in Chimaltenango Department. Trapped sterile flies were discarded, wild adults were counted, washed with sterile water, and placed in a humid chamber (Petri dish with moistened filter paper). After 6 days, the adults were inspected under a stereoscopic microscope in order to verify the presence of fungal mycelium and confirm the positive cases of infection. Infection was calculated as the percentage of wild flies with presence of mycosis.

Data analysis

The survival of *C. capitata* after inoculation was compared among strains with a log-rank test and the MST per replicate was calculated. Two-way ANOVA was used to compare MST among strains. In order to determine the LC_{50} of each strain, the mortality data in each concentration were corrected according to the natural death rate recorded in the control using Abbott's correction (Abbott, 1925). These data were subsequently analyzed using a Probit model (Pc Probit; SAS Institute, 2002). The LC_{50} values obtained from the three strains were subjected to ANOVA. Separation of means was conducted with the Tukey test ($\alpha = 0.05$) (SAS Institute, 2002).

Percentages of infection were arcsine $\sqrt{x}$ -transformed before subjection to ANOVA with repeated measures. The sporulation data per sex were analyzed with a χ^2 test and odds ratio. All analyses were performed using the JMP package (SAS Institute, 2002) with a 95% level of significance. For field evaluation and for each treatment, the weekly data on captured Mediterranean fruit flies were transformed to number of flies per trap per day (FTD). To homogenize the natural variability in population densities (following Maniania et al., 2006), the capture index (I) was estimated according to:

$$I_{ik} = \frac{C_{ik}}{C_{i0}},$$

where C_{ik} is the FTD during week k in treatment i , and C_{i0} is the FTD during the 1st week. Population reduction (%) in each treatment relative to the control was calculated weekly, as:

$$R_{ik} = 100 - \frac{100 \times I_{ik}}{I_{ck}},$$

where R_{ik} is the % reduction in the population in treatment i in week k , I_{ik} is the capture index in any treatment, and I_{ck} is the capture index for the control. The purpose of I_{ik} and R_{ik} was to standardize capture and eliminate apparent differences due to population size per plot. Assuming a model of exponential growth:

$$C_{ik} = C_{i0} \exp(r_{ik}k),$$

where r_{ik} , the intrinsic rate of weekly population growth in treatment i in week k , is obtained as:

$$r_{ik} = \frac{\ln C_{ik} - \ln C_{i0}}{k}.$$

Capture index (I_{ik}), % reduction (R_{ik}), and growth rate (r_{ik}) were plotted over the weeks to explore trends and to compare treatments.

Results

Viability of conidia and median survival time (MST)

Viability of the conidia was acceptable for conducting the bioassays and did not differ between strains: 95% for Bb-ET, 92% for Botanigard, and 90% for Bb-AES ($F_{2,9} = 1.55$, $P = 0.26$). The three strains presented similar biological activity against males of *C. capitata*, producing a MST of 3.8 ± 0.1 , 4.1 ± 0.1 , and 4.2 ± 0.1 days for the strains Bb-ET, GHA, and Bb-AES, respectively ($F_{2,8} = 3.20$, $P = 0.09$) (Table 2).

Mean lethal concentration (LC₅₀)

The mean lethal concentrations (LC₅₀) differed among the evaluated strains, with values of 1.4×10^7 , 2.4×10^6 , and 8.2×10^5 conidia ml⁻¹ for the strains GHA, Bb-AES, and Bb-ET, respectively ($F_{2,9} = 134.9$, $P < 0.001$; Table 3).

Evaluation of sterile male fly vectors and the disseminator devices

The fly capture index decreased with time in the various treatments, but this was more marked in the treatment with sterile male vectors, whereas in the control the index increased (Figure 2). The total numbers of wild-caught flies were 5 452, 1 699, and 1 887 in the treatments with sterile adult vectors, cylindrical device, and panel device, respectively – the percentages of sporulated flies were 57.3, 47.8, and 44.3, respectively (Figure 3). These values did not differ between sterile male vectors and devices ($F_{2,1} = 6.8$, $P = 0.26$). In both devices sporulation (%) did not differ between females and males (cylinder device: $\chi^2 = 0.1$, d.f. = 1, $P = 0.93$; panel device: $\chi^2 = 0.2$, d.f. = 1, $P = 0.65$). In the sterile male vectors treatment, % sporulation was greater in wild males than in females ($\chi^2 = 13.8$, d.f. = 1, $P < 0.001$). In the control, a total of 2 516 adults were captured and no Mediterranean fruit fly showed signs of infection by the fungus.

Table 2 Mortality and median ($\pm$ SE) survival time (MST) of male *Ceratitis capitata* individuals (strain TSL) treated with a 1% suspension of conidia of three formulations of *Beauveria bassiana* (n = 1 000 males per strain)

Strain	Concentration (conidia ml ⁻¹)	Mortality (%)	MST (days)
Bb-ET	2.9×10^7	99.6	3.8 ± 0.1
GHA-Botanigard	4.4×10^8	100	4.1 ± 0.1
22WP			
Bb-AES	7.2×10^7	100	4.2 ± 0.1

Values of MST were not significantly different (two-way ANOVA: $P > 0.05$).

Table 3 Pathogenicity of three formulations of *Beauveria bassiana* against sterile males of *Ceratitis capitata* in laboratory bioassays (n = 2 400 sterile males per strain)

Strains	Replicates	LC ₅₀ (conidia ml ⁻¹)	95% limits of LC ₅₀	LC ₉₀ (conidia ml ⁻¹)	χ^2	P	Equation of regression line ¹
GHA-Botanigard	4	1.4×10^7 a	$1.1 \times 10^7 - 1.6 \times 10^7$	6.4×10^7	11.18	0.090	$y = 0.83x - 13.76$
22WP							
Bb-AES	4	2.4×10^6 b	$2.0 \times 10^6 - 2.8 \times 10^6$	1.4×10^7	6.86	0.095	$y = 0.74x - 10.95$
Bb-ET	4	8.2×10^5 c	$5.9 \times 10^5 - 1.0 \times 10^6$	1.6×10^7	8.38	0.066	$y = 0.42x - 5.82$

¹y = probit, x = log concentration of *Beauveria bassiana* (ln conidia ml⁻¹).

LC₅₀ values followed by different letters are significantly different (ANOVA followed by Tukey tests: $P < 0.05$).

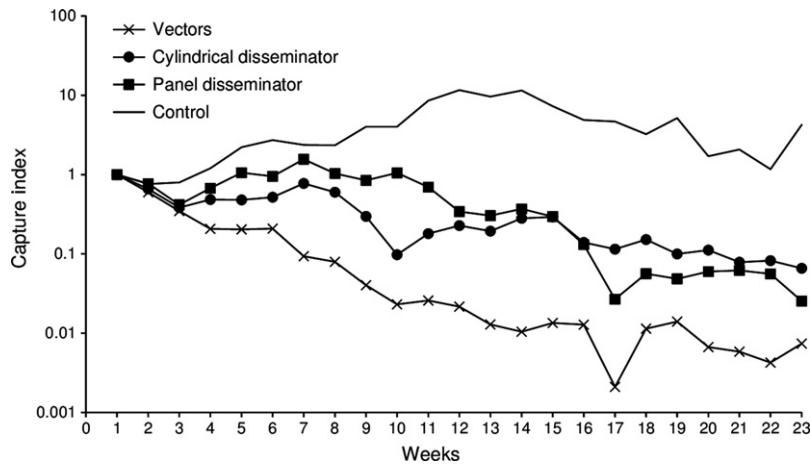


Figure 2 Capture index (note the logarithmic scale) of wild Mediterranean fruit fly in control areas and areas treated with different strategies for disseminating *Beauveria bassiana*.

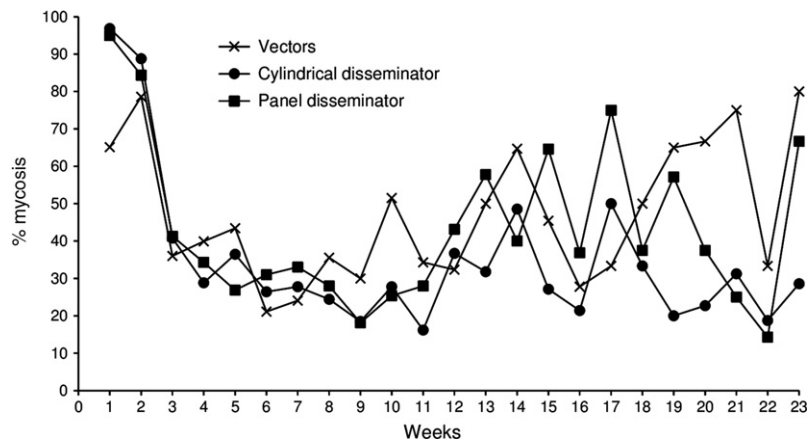


Figure 3 Fungal infection (%) by *Beauveria bassiana* in wild Mediterranean fruit fly populations under different strategies of control.

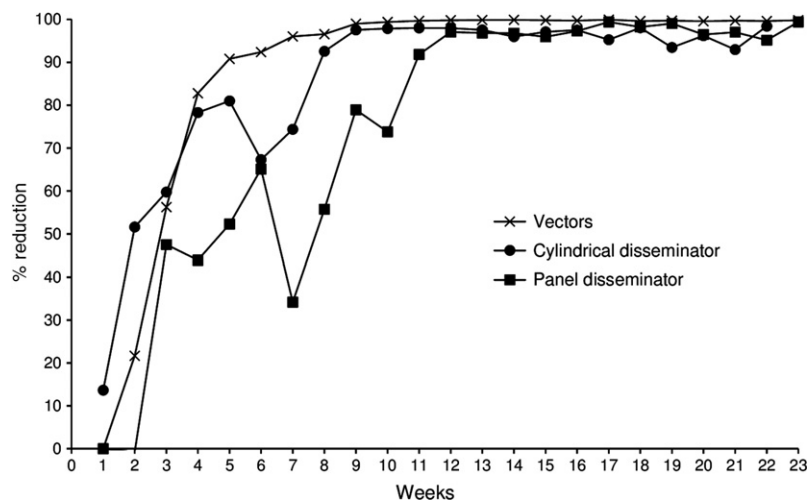
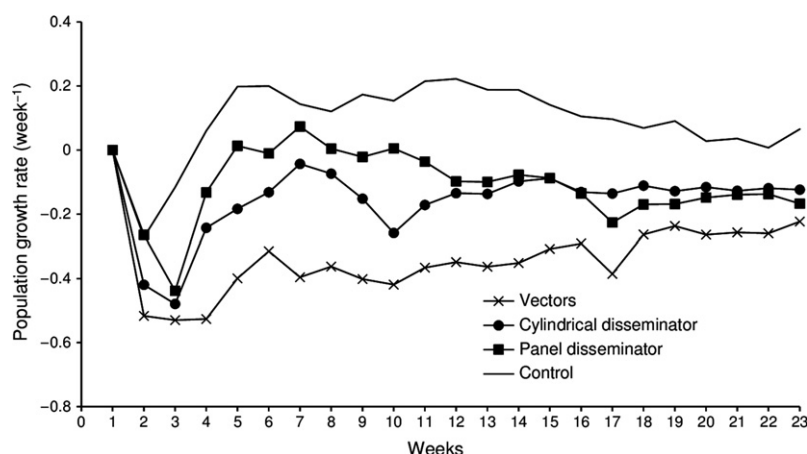


Figure 4 Reduction (%) in wild Mediterranean fruit fly populations in areas under different strategies to disseminate *Beauveria bassiana*, relative to control treatment area.

Wild fly trap captures indicated that the reduction in the population at the end of the study period was >90% for the three treatments. With vectors, the reduction

occurred fast, whereas with the disseminator devices the effect on population reduction was gradual during the 1st weeks (Figure 4). The intrinsic growth rate of the

Figure 5 Intrinsic population growth rate of wild Mediterranean fruit fly in treated and untreated areas under different strategies to disseminate *Beauveria bassiana*.



population was higher in the control treatment, whereas the lowest (and negative) values were observed in the sterile male vectors treatment. The growth rate in the panel devices plots was greater than in the cylinder devices during the 1st weeks (Figure 5).

Discussion

The three strains of *B. bassiana* evaluated against sterile males of the Mediterranean fruit fly displayed high biological activity against the pest, based on the MTS (Quesada-Moraga et al., 2008). Because not all strains of entomopathogenic fungi effectively cause infection in adult fruit flies (De la Rosa et al., 2002; Muñoz et al., 2009), it is necessary to characterize their virulence in bioassays prior to testing them under field conditions. Some strains of *B. bassiana*, evaluated against adults of *C. capitata*, *Ceratitis cosyra* (Walker), and *Ceratitis rosa* (Karsch), presented mortality rates similar to the ones in this study (Dimbi et al., 2003).

The MST determined in this study is considered suitable for strategies of using sterile flies as vectors and for the use of autoinoculation devices, as they allow the vector to live for a longer time and disseminate the inoculum among the wild individuals. Higher virulence would reduce chances for horizontal transmission and will lessen the efficiency of this proposed approach for the control of pest populations in orchard or land with difficult access.

In this study, high levels of population reduction were achieved that were similar to those obtained by Maniania et al. (2006) against the tsetse fly. This is important to pest control programs, as it is possible to achieve high levels of population suppression prior to the release of sterile insects favoring the sterile-to-fertile ratios and sterility induction to the wild population. Releasing sterile insects alone would require higher densities to achieve

adequate sterile-to-fertile ratios, sometimes at a prohibitive cost, compared to the combined more cost-effective approach of population suppression by means of entomopathogens followed by the release of sterile insects (Shelly & McInnis, 2016). The release of sterile male vectors was more effective than the autoinoculation devices in terms of transmitting the conidia to the wild population. This supports previously reported results (Flores et al., 2013) and can be explained by the complex sexual behavior of male aggregations (leks) exhibited by this species (Arita & Kaneshiro, 1985, 1989). Kaneshiro (1993) reported that there is 80% rejection of sterile males by wild females of *C. capitata*, which is evidence of strong sexual selection that could compromise the efficacy of SIT. This defense mechanism may be compensated by the presence of inoculated sterile males, as these can transmit the conidia and cause infection in the wild females even though their attempts at copulation may be rejected (Toledo et al., 2007; Novelo-Rincón, 2009; Flores et al., 2013). The presence of *B. bassiana* spores on the male's body can compensate for the lower sexual performance (Toledo et al., 2007; Dimbi et al., 2009). It has been shown that there are high infections by horizontal transmission of fungal infection from infected flies to healthy ones through mating or physical contact, in small cages or field cages (Toledo et al., 2007; Dimbi et al., 2013). Here, we found that there was horizontal transmission to wild flies under open field conditions. Fungal infection under field conditions was 57.3, 44.7, and 44.3% with sterile fly vectors, panel devices, and cylinder devices, respectively. As these values of infection were found in captured wild flies, a fraction of the population that was infected may have died before being attracted to the trap, so the sporulation levels are likely higher for the total population. This indicates the potential of these methods for successful IPM.

Under natural conditions, *C. capitata* males aggregate in preferred sites such as coffee plantations (Arita & Kaneshiro, 1989; Whittier et al., 1992). The use of conspecifics (whether sterile inoculated males or wild males infected by the disseminator devices) in these sites will increase the possibility of disseminating the inoculum and cause a greater initial suppression of the wild population. This new approach to disperse fungal conidia among conspecifics is simple, economical, and effective, and it has been proposed against the tsetse fly (*Glossinia* spec.) (Maniania & Ekesi, 2013) and other species of fruit flies (Toledo et al., 2007; Sookar et al., 2014; Navarro-Llopis et al., 2015). Population suppression is based on horizontal transmission between wild flies and fungal inoculated males, using disseminator devices or sterile male vectors (Dimbi et al., 2009; Navarro-Llopis et al., 2015). In addition, the percentage of infection in wild females and males did not differ, which suggests that wild males attracted to the devices baited with trimedlure were capable of transmitting the inoculum during interaction with the wild females. The use of disseminator devices may be a more suitable control method in situations where there is no infrastructure to support the massive production and release of sterile flies to use as vectors.

With the use of sterile flies as vectors and disseminator devices, fruit flies could be controlled with greater efficacy, mainly in reservoirs of pest populations or in organic cultivation systems where other more conventional control methods cannot be applied. It has been reported that *B. bassiana* is a generalist fungus that can affect a wide range of insects (Goettel et al., 1990; Suckling et al., 2012). However, previous studies have demonstrated that the release of sterile flies as vectors in large areas is highly specific and did not cause infection in bees or coffee berry borers (*Hypothenemus hampei* Ferrari) present in the area (Flores et al., 2013). One additional advantage is that only trace quantities of conidia are required in order to inoculate sterile flies (0.0001 g per fly). For this reason the additional cost when used in combination with SIT is low if compared with the benefit obtained from the complementary effect of using both techniques for suppressing and eradicating the pest. The combined control methods would be particularly useful in sites with difficult access, organic agricultural systems, backyard hosts, and natural protected areas where insecticide applications are not allowed. It would also be useful for fruit fly population control during the rainy season, when the use of ground and aerial insecticide sprays becomes ineffective.

In conclusion, this study proposes three novel methods for autodissemination of *B. bassiana* conidia which constitute viable environmentally friendly control methods

that can be incorporated into strategies for fruit fly control, including those that integrate SIT against the Mediterranean fruit fly.

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